Finite element solution of the stationary Schrödinger eigenvalue problem: convergence, sparse eigensolution strategies, and code-to-code verification

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Abstract

We present and analyse a finite element framework for the stationary Schrödinger eigenvalue problem in one to three spatial dimensions. The weak formulation yields a generalized symmetric eigenproblem $\boldsymbol{A\psi = EM\psi}$ with $\boldsymbol{A = K + M_V}$. We establish two properties of the discrete spectrum: when the discrete form is the exact restriction of the continuous one, the conforming approximation produces eigenvalues that are upper bounds for the exact ones, and under standard regularity assumptions the eigenvalue error converges as $\boldsymbol{\mathcal{O}(h^{2s})}$. The implementation replaces the potential by its nodal interpolant, so the upper-bound property is preserved for convex $\boldsymbol{V}$ and is verified numerically otherwise rather than assumed. We show that the usual diagonal-shift treatment of Dirichlet data introduces spurious eigenvalues, whereas elimination of the constrained degrees of freedom reproduces the constrained spectrum exactly. Three eigensolution strategies are considered. Two—dense factorization and sparse shift-and-invert Arnoldi iteration—are compared experimentally on identical matrices; the third, a distributed solution, is analysed and its deployment requirements and present limitations are documented, but it is not benchmarked here. A cost model separating assembly, factorization, iteration and solver overhead accompanies the comparison. On measured benchmarks the sparse strategy is up to $\mathbf{4097}\times$ faster, reduces storage from $\boldsymbol{\mathcal{O}(n^2)}$ to $\boldsymbol{\mathcal{O}(\mathrm{nnz})}$, and renders a problem with $\mathbf{3.96 \times 10^4}$ degrees of freedom infeasible in dense arithmetic ($\mathbf{37.6}$ GB) solvable in $\mathbf{0.41}$ s. Verification is fourfold: against closed-form solutions; by code-to-code comparison of two independently written implementations, agreeing to about $\mathbf{10^{-11}}$; by

structural checks; and by confirming the predicted $\mathcal{O}(h^2)$ convergence. We also document a limitation of the in-framework solver under domain decomposition.

Keywords: Schrödinger equation · Finite element method · Generalized eigenvalue problem · Sparse eigensolvers · Shift-and-invert Arnoldi · Code verification

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1 Introduction

From a mathematical point of view, quantum mechanics can be regarded in many respects as an eigenvalue problem: one has to compute eigenvalues and eigenvectors of Hamiltonians, and the knowledge of the ground state and of the first few excited states already yields considerable insight into the physics of a system [1]. The stationary Schrödinger equation is the canonical instance [2, 3]. In dimensionless form it reads

$$-\nabla^2\psi(\mathbf{r}) + V(\mathbf{r})\,\psi(\mathbf{r}) = E\,\psi(\mathbf{r}), \qquad \mathbf{r} \in \Omega \subset \mathbb{R}^d, \quad d = 1, 2, 3, \tag{1}$$

where $\psi = \psi(\mathbf{r})$ is the wavefunction, $V = V(\mathbf{r})$ the confining potential, E the energy of the state and Ω the domain, with $\partial\Omega$ its boundary. Equation (1) is subject to boundary conditions on $\partial\Omega$.

Closed-form solutions exist only for a small number of separable potentials—the infinite well, the harmonic oscillator, the Coulomb problem—while the problems of practical interest require numerical treatment. In semiconductor physics the relevant structures are quantum wells, wires and dots, heterostructures and resonant tunnelling diodes, where quantum confinement modifies the electronic structure and hence the optical and transport properties [4, 5]. In these applications the equation is usually written in the effective-mass form

$$-\frac{\hbar^2}{2m^*}\nabla^2\psi + V\psi = E\psi, \tag{2}$$

where $\hbar$ is the reduced Planck constant and m^* the carrier effective mass, both absorbed into the unit coefficient of (1). The relation between this form and (1) is made explicit in Section 2.2, since it controls the interpretation of all reported eigenvalues.

Two families of numerical methods dominate the literature. Finite difference discretizations are simple and yield banded or sparse matrices, but represent curved or irregular geometries poorly. Finite element methods handle complex geometries and unstructured meshes naturally, and their convergence theory for eigenvalue problems is well established [6–8]. Discontinuous Galerkin variants have been developed specifically for the Schrödinger equation in device contexts, including multiscale and WKB-based formulations of the stationary Schrödinger–Poisson problem [9, 10]. Finite element approaches to quantum systems have a long history, from low-dimensional models and quantum dots [1, 5] to quantum transport in nanoscale devices [3, 11],

where the eigenproblem is typically coupled to a Poisson equation. The wider context of this work is the multiphysics simulation of power apparatus, in which coupled electromagnetic, thermal and structural fields are already solved within a single finite element model [12]; a quantum-mechanical eigenproblem solver extends that class of models to problems where carrier confinement must be represented.

A practical difficulty rarely discussed is that of *solution strategy*. A finite element discretization of (1) with n unknowns produces matrices that are sparse, yet many implementations—including the one we started from—assemble them densely and call a dense LAPACK driver. This is adequate for a few thousand unknowns and hopeless beyond that: storage grows as $\mathcal{O}(n^2)$ and time as $\mathcal{O}(n^3)$. Meanwhile only the lowest few eigenpairs are usually of physical interest, which is precisely the regime in which iterative Krylov methods with shift-and-invert spectral transformation are effective [13, 14]. The inner solves of such methods rely on sparse direct factorizations [15] whose availability in parallel determines whether the scheme can be distributed. The choice between dense, sparse and distributed solution is therefore not a detail of implementation but a modelling decision, and we treat it as such.

A third difficulty, of a different nature, is that of *trust*: how does one establish that an implementation solves the equations it is intended to solve? Comparison against analytic solutions at a single resolution is weak evidence, since it cannot distinguish compensating errors. A stronger approach, well established in the verification and validation literature [16, 17], is cross-checking against an independently written implementation. The method of manufactured solutions [18, 19] provides a complementary route when exact solutions are unavailable. Such cross-checks do appear in the computational physics literature—an early example being a comparison of independent particle-in-cell codes [20]—but they remain uncommon; we adopt that route here, in the form of the code-to-code comparison of Section 7.4.

Contributions

The work introduces no new numerical algorithm: the elements, quadrature rules and eigensolvers are standard, and we state this plainly. The contributions are instead the following.

1. A dimension-generic finite element formulation for (1) in one, two and three dimensions, supporting arbitrary scalar potentials in one dimension and separable potentials, $V = \sum_k V_k(x_k)$, in higher dimensions (Section 6.3), implemented as a module of the open-source finite element package Elmer [21], to our knowledge the first such module for this equation in that framework; we make no claim about other packages.
2. Two structural results on the discrete spectrum (Section 4): the conforming eigenvalues are upper bounds for the exact ones, and the eigenvalue error obeys the classical $\mathcal{O}(h^{2s})$ estimate. The predicted order is verified numerically.
3. A quantified comparison of the dense and sparse eigensolution strategies on identical matrices, with a cost model that predicts the observed $\mathcal{O}(n^2)$ growth of the speed-up (Sections 5 and 8.3).

4. A four-way verification programme (Section 7) combining analytic benchmarks, code-to-code comparison of two independent implementations, structural consistency checks, and convergence-order confirmation.
5. Documentation of two failure modes that are easy to encounter and hard to diagnose: a tensor-product assembly error that is invisible in one dimension (Remark 2), and silently incorrect results when the in-framework solver is run under domain decomposition (Section 8.4).

The remainder of the paper is organised as follows. Section 2 states the problem, derives the weak form and discusses the unit convention. Section 3 describes the discretization and the treatment of Dirichlet data. Section 4 establishes the properties of the discrete spectrum. Section 5 analyses the eigensolution strategies. Section 6 describes the two implementations. Section 7 presents the verification programme and Section 8 the numerical experiments. Section 10 concludes.

2 Problem setting and notation

2.1 Strong form

Let $\Omega \subset \mathbb{R}^d$ be a bounded, open, connected domain with Lipschitz boundary. We seek a pair (ψ, E), with $\psi \neq 0$, satisfying the stationary Schrödinger eigenvalue problem [2, 3]:

$$-\nabla^2\psi + V\psi = E\psi \quad \text{in } \Omega, \qquad \psi = 0 \quad \text{on } \partial\Omega. \tag{3}$$

It is worth distinguishing three problems that are conflated in much of the literature, because only the first is what the discrete equations actually solve.

1. **The bounded-domain problem.** Ω is the box of (3) and the Dirichlet condition is part of the statement. This is the problem the finite element method approximates, and for it $H^1_0(\Omega)$ embeds compactly into $L^2(\Omega)$, so the spectrum is discrete and bounded below with no further hypothesis on V.
2. **The unbounded problem.** $\Omega = \mathbb{R}^d$ and discreteness must be earned by the potential: we assume throughout that V is confining, $V(\mathbf{r}) \to \infty$ as $|\mathbf{r}| \to \infty$, so that the resolvent is again compact and $E_1 \leq E_2 \leq \cdots \to \infty$ with finite multiplicities. This is the physical reference problem, and the analytic values quoted in Table 1 belong to it.
3. **The truncated and discretized problem.** What is computed: problem 1 on a mesh of size h in a box of side L.

The passage from 2 to 3 carries a *truncation* error, controlled by L, and the passage from 3 to the discrete equations carries a *discretization* error, controlled by h. The two are of different nature and are separated numerically in Section 8.2: there the mesh is held fixed while L varies, and vice versa. Where a case has a closed-form solution on the half-line rather than in a box—the Morse potential is the example here—the comparison carries the truncation error as well, and is quoted with that caveat.

2.2 Units and the effective-mass interpretation

Equation (3) places a unit coefficient on the Laplacian. We now make the physical content of that choice explicit, since it controls how results should be interpreted.

Equations (2) and (3) differ only by the coefficient $\hbar^2/2m^*$ multiplying the Laplacian [3, 4]. Let L be a characteristic length of the structure [4] and set

$$\mathcal{E} = \frac{\hbar^2}{2m^* L^2}, \qquad \mathbf{r} = L\,\tilde{\mathbf{r}}, \tag{4}$$

so that $\mathcal{E}$ is the energy unit associated with L and $\tilde{\mathbf{r}}$ is the position vector in units of L. We use the calligraphic symbol for the unit deliberately, reserving E_0 for the ground-state energy and E_n for the n-th eigenvalue in increasing order, as in Appendix A. Dividing (2) by $\mathcal{E}$ and using $\nabla^2 = L^{-2}\tilde{\nabla}^2$ gives, after dropping tildes,

$$-\tilde{\nabla}^2\psi + \frac{V}{\mathcal{E}}\psi = \frac{E}{\mathcal{E}}\psi, \tag{5}$$

which is (3) with $\tilde{V} = V/\mathcal{E}$ and $\tilde{E} = E/\mathcal{E}$.

Remark 1 (Physical scaling) Equation (3) is the effective-mass equation in units where $\hbar^2/2m^* = 1$. Eigenvalues reported here are dimensionless and become physical energies upon multiplication by $\mathcal{E}$. For GaAs, where $m^* \approx 0.067\,m_e$, and a nominal length $L = 10\,\text{nm}$, one finds $\mathcal{E} \approx 5.7\,\text{meV}$. Thus the two lowest levels of the harmonic potential $V = x^2$ in (3), namely $E_1 = 1$ and $E_2 = 3$, correspond to 5.7 and 17.1 meV at that length scale. Quadratic confinement of this form is the standard model for a parabolic quantum well [4, 22], and shifts of a few meV in the confined levels are precisely what controls intersubband optical transitions in such structures, as well as the subband structure of modulation-doped wells [23].

Consequently, when the conventional form $-\frac{1}{2}\nabla^2\psi + \frac{1}{2}\omega^2 r^2\psi = E\psi$ (units $\hbar = m = 1$) is used, multiplying by two gives $-\nabla^2\psi + \omega^2 r^2\psi = 2E\psi$; comparison with (3) shows that a potential $V = \alpha r^2$ corresponds to $\omega = \sqrt{\alpha}$, and that the conventional energies are half of those reported here. All analytic references are stated in the convention of (3) (Appendix A) to avoid ambiguity.

2.3 Weak form and variational characterisation

Let $H_0^1(\Omega) = \{v \in H^1(\Omega) : v|_{\partial\Omega} = 0\}$. Multiplying (3) by $v \in H_0^1(\Omega)$ and integrating by parts [6, 8],

$$\underbrace{\int_\Omega \nabla\psi \cdot \nabla v\,\mathrm{d}\Omega}_{a(\psi,v)} + \underbrace{\int_\Omega V\psi v\,\mathrm{d}\Omega}_{b_V(\psi,v)} = E\,\underbrace{\int_\Omega \psi v\,\mathrm{d}\Omega}_{m(\psi,v)}. \tag{6}$$

The forms a, b_V and m are symmetric, and m is positive definite on $H_0^1(\Omega)$. The form a is H_0^1-elliptic, but b_V is *not* elliptic when V takes negative values, as it does in several of the benchmark cases below ($V_{\min} = -10$ for Morse and $-1/2$ for the

sinusoidal potential). What is required is a hypothesis on the full energy form

$$q(u,v) = a(u,v) + b_V(u,v) = \int_\Omega \big(\nabla u \cdot \nabla v + V\,uv\big)\,\mathrm{d}\Omega. \tag{7}$$

We assume throughout that $V \in L^\infty(\Omega)$ is bounded below, so that b_V is bounded and q is bounded below on $H^1_0(\Omega)$; by the theory of closed semibounded forms [2, 8], the form $q + c\,m$ is continuous and coercive for c large enough, so it defines a unique self-adjoint operator H with $\langle Hv, v\rangle = q(v,v)$ on its domain; the shift by c is a bounded perturbation and does not affect discreteness. Because Ω is bounded, $H^1_0(\Omega)$ embeds compactly into $L^2(\Omega)$, so the resolvent of H is compact. Its spectrum is therefore discrete and bounded below, and consists of an increasing sequence $E_1 \le E_2 \le \cdots \to \infty$ of finite multiplicity. *The eigenvalues need not be positive*: for the Morse potential all four reported levels are negative (Table 6). The ordering is by value, not by sign, and the Rayleigh quotient (8) must be replaced by $q(v,v)/m(v,v)$, which is minimised in the same way. Defining the Rayleigh quotient [2]

$$R(v) = \frac{a(v,v) + b_V(v,v)}{m(v,v)}, \qquad v \in H^1_0(\Omega)\setminus\{0\}, \tag{8}$$

the eigenvalues admit the min–max characterisation [2, 7, 8]

$$E_n = \min_{\substack{S_n \subset H^1_0(\Omega)\\ \dim S_n = n}} \; \max_{0 \neq v \in S_n} R(v). \tag{9}$$

This characterisation is the basis of the results in Section 4.

3 Finite element discretization

3.1 Multilinear elements and quadrature

Let $\mathcal{T}_h$ be a conforming partition of Ω into d-linear elements—intervals, quadrilaterals or hexahedra—with mesh size h. We denote by $\tau_e \in \mathcal{T}_h$ a generic element. The discrete space is

$$V_h = \{v_h \in H^1_0(\Omega) : v_h|_{\tau_e} \in \mathcal{Q}_1(\tau_e) \;\; \forall \tau_e \in \mathcal{T}_h\} \subset H^1_0(\Omega), \tag{10}$$

where $\mathcal{Q}_1$ denotes multilinear polynomials [6]. The inclusion $V_h \subset H^1_0(\Omega)$ makes the method conforming and is used in Theorem 1. On each element the approximation is $\psi_h|_{\tau_e} = \sum_{i=1}^{2^d} N_i^{(e)}(\boldsymbol{\xi})\,\psi_i^{(e)}$.

Volume integrals are evaluated with 2^d-point Gauss–Legendre quadrature per element [6], and for each element we compute

$$K_{ij}^{(e)} = \int_{\tau_e} \nabla N_i \cdot \nabla N_j \,\mathrm{d}\Omega, \quad M_{ij}^{(e)} = \int_{\tau_e} N_i N_j \,\mathrm{d}\Omega, \quad M_{V,ij}^{(e)} = \int_{\tau_e} N_i V N_j \,\mathrm{d}\Omega. \tag{11}$$

Remark 2 (The tensor-product trap) On a tensor-product mesh the multidimensional matrices are *sums* of Kronecker products in which exactly one factor is a 1D stiffness matrix and the remaining factors are 1D mass matrices. In two dimensions,

$$K = K_1 \otimes M_1 + M_1 \otimes K_1, \qquad M = M_1 \otimes M_1, \tag{12}$$

and in three dimensions K is the sum of the three cyclic terms $K_1 \otimes M_1 \otimes M_1 + M_1 \otimes K_1 \otimes M_1 + M_1 \otimes M_1 \otimes K_1$, with $M = M_1 \otimes M_1 \otimes M_1$. A tempting shortcut is to use a single Kronecker product of 1D stiffness matrices, $K_1 \otimes K_1$ in two dimensions. That is *incorrect*: it omits the cross terms in which the derivative acts on only one of the two coordinates, and it is not even dimensionally consistent, since K_1 scales as h^{-1} while the multidimensional stiffness scales as h^{d-2}. The same remark applies to the potential term, which for a separable V is again a sum of terms with one weighted mass matrix $\int N_i V_k N_j$ in the corresponding slot. During development this error produced 2D and 3D eigenvalues roughly 20× too small while leaving the 1D results correct—a signature that made the defect hard to attribute, since the one-dimensional verification passed. Assembling (11) by direct quadrature for every dimension removes the error. We report it because it is easy to introduce in dimension-generic codes and produces results that look plausible in the dimension where they are first tested.

3.2 Assembly and the generalized eigenproblem

Assembling (11) over $\mathcal{T}_h$ gives global matrices $K, M, M_V \in \mathbb{R}^{n\times n}$, symmetric, with K and M_V sparse and

$$A = K + M_V, \qquad A\,\boldsymbol{\psi} = E\,M\,\boldsymbol{\psi}. \tag{13}$$

This is a generalized symmetric eigenproblem with M symmetric positive definite [6, 14]. Its eigenvectors are M-orthonormal: $\boldsymbol{\psi}_i^{\top} M \boldsymbol{\psi}_j = \delta_{ij}$, which provides an implementation-independent check used in Section 7.5.

3.3 Dirichlet boundary conditions

The treatment of constrained degrees of freedom deserves attention, because the approach most common for boundary value problems is incorrect here, and the way in which it fails depends on how the mass matrix is treated.

In a source problem $Au = f$, Dirichlet data are frequently imposed by zeroing the row and column of A and placing 1 on the diagonal. Two variants of this recipe appear in practice when it is carried over to (13).

1. $A_{ii} = 1$ *and* $M_{ii} = 1$ for a constrained node i. The pair (A_{ii}, M_{ii}) then defines a one-dimensional generalized problem whose Rayleigh quotient is exactly 1, so the discrete spectrum contains $E = 1$ with an eigenvector supported on the constrained nodes. With n_c constrained nodes the value 1 appears with multiplicity up to n_c, independently of the physical problem.
2. $A_{ii} = 1$ with M_{ii} left at its assembled value. The diagonal quotient is then $1/M_{ii} \neq 1$, and no eigenvalue is pinned at 1; but the constrained rows no longer represent the variational problem, and the spectrum is perturbed in a way that is largest for the low-lying states of interest.

We verified both variants numerically on the 1D harmonic case ($L = 20$, 40 elements). With $A_{ii} = M_{ii} = 1$ the two lowest computed eigenvalues are 1.000000 and

1.000000, against the exact 1 and 3: the low-frequency spectrum is dominated by two spurious modes. With $A_{ii} = 1$ and M_{ii} unchanged the spurious value disappears, but the two lowest eigenvalues become 1.057052 and 3.117594, which are also wrong. In either case the constrained rows must be *removed*, not modified.

The correct procedure eliminates the constrained unknowns [6]. The following statement makes the equivalence precise.

Proposition 1 (Elimination) *Let W_h be the finite element space of multilinear functions on the mesh,* before *any boundary condition is imposed, and let $V_h = W_h \cap H_0^1(\Omega)$ be the conforming subspace obtained by requiring that the nodal values on $\partial\Omega$ vanish. Write V_h^0 for the further subspace of V_h in which the constrained nodal values are set to zero, so that V_h^0 is the discrete counterpart of H_0^1 once the degrees of freedom attached to the constrained nodes have been removed. (In the mesh used here the constrained nodes are exactly those on $\partial\Omega$, so $V_h^0 = V_h$ as sets of functions; the distinction matters when constrained nodes are interior, for instance on an interface.) Then the variational eigenproblem on V_h^0,*

$$q(\psi_h, v_h) = E_h\, m(\psi_h, v_h) \qquad \forall v_h \in V_h^0, \tag{14}$$

is equivalent to the reduced algebraic problem

$$A_{ff}\,\boldsymbol{\psi}_f = E\, M_{ff}\,\boldsymbol{\psi}_f : \tag{15}$$

the two have the same eigenvalues, and the nodal coefficients of an eigenfunction of (14) are $(\boldsymbol{\psi}_f, \mathbf{0})$. Moreover A_{ff} and M_{ff} are symmetric and M_{ff} is positive definite.

Proof The basis functions φ_j with $j \in f$ span V_h^0, since the coefficients of the constrained basis functions vanish there. Writing $v_h = \sum_{j\in f} v_j \varphi_j$ and testing (14) with φ_i, $i \in f$, gives

$$\sum_{j\in f} (A_{ij} - E_h M_{ij}) v_j = 0 \qquad \forall i \in f, \tag{16}$$

which is precisely (15) with $A_{ff} = (A_{ij})_{i,j\in f}$ and $M_{ff} = (M_{ij})_{i,j\in f}$. Conversely, every solution of (15), extended by zero on the constrained nodes, satisfies (14), because the φ_j with $j \in f$ span V_h^0. The block rows indexed by c are *not* imposed, and are in general not satisfied: eliminating a degree of freedom means discarding its equation, not merely setting its coefficient to zero (Remark 3). Symmetry of A_{ff} and M_{ff} follows by restriction. For positive definiteness, let $\boldsymbol{\psi}_f \neq \mathbf{0}$; then $\boldsymbol{\psi}_f^{\top} M_{ff} \boldsymbol{\psi}_f = (\boldsymbol{\psi}_f, \mathbf{0})^{\top} M (\boldsymbol{\psi}_f, \mathbf{0}) > 0$, because M is positive definite and $(\boldsymbol{\psi}_f, \mathbf{0}) \neq \mathbf{0}$. □

Remark 3 (The discarded rows are not satisfied) It is tempting to describe the elimination as imposing $\boldsymbol{\psi}_c = \mathbf{0}$ in the block form of (13) and then observing that the first block row reduces to (15). The second block row, however, becomes $A_{cf}\boldsymbol{\psi}_f = E M_{cf}\boldsymbol{\psi}_f$, which is an additional constraint and holds only by accident. We checked this numerically on two benchmark cases. For the 1D harmonic oscillator the residual of the discarded row is 8×10^{-15}, that is, round-off—but only because the exact eigenfunction has already decayed to $\sim 10^{-17}$ at the first interior node. For the infinite well, where the eigenfunction is $\mathcal{O}(h)$ near the wall, the same residual is 1.8×10^{-2} against a peak amplitude of 2.2×10^{-1}, that is, 8%. The distinction is therefore not pedantic: it separates a proof that holds in general from one that holds only on the subclass of problems whose eigenfunctions decay fast enough at the boundary.

Both implementations perform this elimination, and the number of free degrees of freedom n_f is reported for every case so that the effective problem size is unambiguous.

4 Convergence and properties of the discrete spectrum

We now establish two properties of the conforming approximation that serve both as theoretical justification and as practical checks.

4.1 The discrete eigenvalues are upper bounds

Theorem 1 (Upper bound) *Let q be the energy form (7) and suppose $V \in L^\infty(\Omega)$ is bounded below, so that q is bounded below and the continuous problem (6) has a discrete spectrum $E_1 \le E_2 \le \cdots$. Let $E_{h,1} \le E_{h,2} \le \cdots \le E_{h,n_h}$ be the eigenvalues of the discrete problem*

$$q(\psi_h, v_h) = E_h\, m(\psi_h, v_h) \qquad \forall v_h \in V_h, \tag{17}$$

with $n_h = \dim V_h$. If $V_h \subset H_0^1(\Omega)$ and the discrete form is the restriction of q to $V_h \times V_h$, then

$$E_{h,n} \ge E_n \qquad \text{for all } n \le n_h. \tag{18}$$

Proof By (9), $E_n = \min_{\dim S_n = n} \max_{0 \neq v \in S_n} R(v)$ with $S_n \subset H_0^1(\Omega)$, whereas the discrete eigenvalues satisfy the same formula with S_n restricted to subspaces of V_h [2, 7, 8]. Since $V_h \subset H_0^1(\Omega)$, the set of admissible subspaces in the discrete case is a subset of the admissible set in the continuous case, and minimising over a smaller set cannot decrease the minimum. Hence $E_{h,n} \ge E_n$. □

Remark 4 (Scope of the upper bound) The hypothesis that the discrete form be the restriction of q is essential and is *not* satisfied exactly by the implementations used here, which replace V by its nodal interpolant V_h when assembling the potential term (Remark 7). The consequence depends on the sign of $V - V_h$. If V is convex, as for the harmonic potential $V = x^2$, the piecewise-linear interpolant satisfies $V_h \ge V$ pointwise, hence $q_h(v, v) \ge q(v, v)$ for every $v \in V_h$ and the upper-bound property survives. For a general, non-convex V no such inequality holds, and the computed eigenvalues need not be upper bounds for those of the continuous problem. The property is therefore asserted for the exactly integrated problem, and *verified numerically* for the cases at hand (Section 7.5) rather than assumed.

Remark 5 Subject to that scope, Theorem 1 has a practical consequence: computed eigenvalues approach the exact ones from above, so a mesh refinement that moves an eigenvalue in the wrong direction signals an implementation error rather than a discretization effect. We use this as a consistency check in Section 7.5, while noting that monotonicity alone is not a proof of correctness.

4.2 A priori eigenvalue error estimate

Theorem 2 (Convergence) *Let $\{\mathcal{T}_h\}_h$ be a shape-regular family of partitions of Ω into multilinear elements of degree p, and let the discrete form be the restriction of q to $V_h \times V_h$. Suppose that the eigenvalues $E_n, \ldots, E_{n+m-1}$ form a group of multiplicity m separated from*

the rest of the spectrum, and that the associated spectral subspace $\mathcal{E} = \operatorname{span}\{u_n, \dots, u_{n+m-1}\}$ *satisfies* $\mathcal{E} \subset H^{1+s}(\Omega)$ *for some* $s \in (0, p]$*. Then there exist constants* C *and* C'*, depending on the group, on the separation distance and on the shape regularity of* $\mathcal{T}_h$ *but not on* h*, such that*

$$|E_{h,\ell} - E_\ell| \le C\, h^{2s} \qquad \ell = n, \dots, n+m-1, \tag{19}$$

and, for a suitable orthonormal basis $\{\psi_{h,\ell}\}$ *of the discrete spectral subspace,* $\|\psi_{h,\ell} - u_\ell\|_{H^1} \le C' h^s$*.*

The estimate (19) is the classical result of Babuška and Osborn [7, 8], stated here for a spectral group rather than for a simple eigenvalue; the simple case is $m = 1$. Two features deserve emphasis.

First, the exponent s is a property of the *problem*, not of the method: it is limited by the smoothness of $\partial\Omega$, by the regularity of V and by the boundary conditions. On a convex polygon with $V \in L^\infty$ one has $u_n \in H^2$ and $s = 1$; on a domain with re-entrant corners, or for a potential with a discontinuity, $s < 1$ and the asymptotic rate is correspondingly reduced. The estimate also presumes a *shape-regular* mesh family and a consistent approximation of the same variational problem.

Second, the estimate concerns the *spectral subspace*, not individual eigenfunctions: within a group of multiplicity $m > 1$ the individual eigenvectors are not determined by the differential operator, so only the subspace—equivalently, the set of eigenvalues—is a well-posed object. This is the reason the degeneracies of Section 7.5 are a meaningful test, whereas a comparison of individual eigenvectors within a degenerate group is not.

Corollary 3 *Let* Ω *be a bounded convex polygon and let* $V \in L^\infty(\Omega)$ *with* $\mathcal{E} \subset H^2(\Omega)$*. Then for multilinear elements* ($p = 1$*,* $s = 1$*)*

$$|E_{h,\ell} - E_\ell| = \mathcal{O}(h^2). \tag{20}$$

The hypothesis $\mathcal{E} \subset H^2$ is what makes $s = 1$ available. It holds for a convex polygon in two dimensions by the classical elliptic regularity theory [7, 8], and for $V \in L^\infty$ it is not lost, since the potential term is a lower-order perturbation. In three dimensions the statement needs care: H^2 regularity on a general Lipschitz polyhedron is *false*—the solution develops edge and vertex singularities even for smooth data—so Corollary 3 should not be read as a claim about arbitrary hexahedral meshes on a general polyhedron. The boxes used here are rectangular parallelepipeds, for which the solution is as regular as the data permits, and the observed order in Table 8 is 1.98–1.99 in 3D, consistent with $s = 1$. Where the domain is not convex the exponent drops below 1 and the observed order falls accordingly.

Remark 6 (Theoretical versus observed order) The order reported in Tables 2 and 8 is an *observed* quantity: it is obtained as $\log(e_i/e_{i+1})/\log(h_i/h_{i+1})$ from the relative errors of two consecutive refinements, and it is therefore a local estimate that inherits the noise of both. It should be read as evidence consistent with Corollary 3, not as a measurement of the exponent s. The tables report the relative error $|E_{h,1} - E_1|/|E_1|$; the corresponding absolute errors differ by the factor $|E_1|$, which is why the Morse case, with $|E_1| \approx 8.5$, appears to have larger absolute deviations than the infinite well at the same resolution. Convergence

of eigenfunctions is a separate question: the H^1 error is only $\mathcal{O}(h^s)$, and a comparison of eigenfunctions additionally requires a normalisation convention and a choice of sign, both of which are arbitrary.

5 Eigensolution strategies

Only the lowest few eigenpairs of (15) are normally of interest. We consider three strategies and analyse their cost.

5.1 Dense direct solution

The classical approach reduces (15) to a standard eigenproblem via a Cholesky factorization $M_{ff} = LL^{\top}$ and calls the LAPACK driver `DSYGV` [15, 24], which computes *all* eigenpairs. It is extremely robust and requires no tuning. Its cost is the obstacle. Storing the two matrices A_{ff} and M_{ff} in double precision requires at least

$$\mathrm{mem}_{\mathrm{dense}} \;\geq\; 16\, n_f^2 \text{ bytes}, \tag{21}$$

and the factorization costs $\mathcal{O}(n_f^3)$ operations. The implementations used here hold K, M and M_V separately before forming A, so their footprint is $24\, n_f^2$; at $n_f = 4 \times 10^4$ that is 38.4 GB, against 25.6 GB for the irreducible two-matrix minimum (throughout, GB denotes 10^9 bytes and not 2^{30}), in either case before any workspace.

5.2 Sparse shift-and-invert Arnoldi iteration

The alternative exploits sparsity and the fact that only a few eigenpairs are wanted. Implicitly restarted Arnoldi iteration [13] computes the eigenvalues of largest magnitude of a matrix; to obtain the *smallest* eigenvalues of (15) we use the shift-and-invert transformation [25], which converts the problem into one whose dominant eigenvalues correspond to those nearest a prescribed shift σ:

$$(A_{ff} - \sigma M_{ff})^{-1} M_{ff}\, \mathbf{v} = \mu\, \mathbf{v}, \qquad E = \sigma + \frac{1}{\mu}. \tag{22}$$

Here $\mathbf{v}$ is the eigenvector of the transformed problem and the eigenvalues μ of largest magnitude correspond to the eigenvalues E of (15) nearest σ. The factorization of $(A_{ff} - \sigma M_{ff})$ is performed *once*, in sparse form, before the iteration [15], giving

$$\mathrm{mem}_{\mathrm{sparse}} = \mathcal{O}(\mathrm{nnz}) + \mathcal{O}(n_f k), \tag{23}$$

where nnz denotes the number of nonzeros of $(A_{ff} - \sigma M_{ff})$ and k the number of requested eigenpairs. This is the strategy implemented in the Python reference code and is available to the Elmer module through matrix export.

5.3 Distributed solution

The same transformation can be distributed: PETSc [26] provides the parallel sparse matrix infrastructure and SLEPc [27] the Krylov–Schur eigensolver with shift-and-invert. Both rely on a *parallel* sparse direct factorization for the inner solve; without one the shift-and-invert step cannot be distributed and the driver is confined to a single process. This is a deployment constraint rather than an algorithmic one, and we report what we measured in Section 8.3.

5.4 Cost model

For a quasi-uniform mesh in d dimensions each node couples to $\mathcal{O}(1)$ neighbours, so the number of nonzeros of A_{ff} and M_{ff} grows linearly, $\mathrm{nnz} = \mathcal{O}(n_f)$, and the storage estimate (23) holds. The *time*, however, is not captured by a single term, and we prefer to separate the contributions rather than to state a universal law.

A shift-and-invert Arnoldi solve consists of:

1. the assembly of A_{ff} and M_{ff}, of cost $\mathcal{O}(\mathrm{nnz})$;
2. one sparse direct factorization of $(A_{ff} - \sigma M_{ff})$, of cost $\mathcal{O}(\mathrm{nnz}^{\alpha})$ with $1 < \alpha \leq 3/2$ in two and three dimensions depending on the fill-in, the reordering and the elimination tree [15];
3. n_{it} Arnoldi iterations, each requiring one back-substitution with the stored factors, of cost $\mathcal{O}(\mathrm{nnz}_L)$ where nnz_L is the number of nonzeros of the factor;
4. the eigensolver overhead proper, of order $n_f k^2$ for k vectors of length n_f.

The factorization cost is therefore *not* linear in n_f in general, and in the regime studied here it is the second term that dominates. Writing the four contributions separately,

$$T_{\mathrm{sparse}} \sim \underbrace{c_1\,\mathrm{nnz}}_{\text{assembly}} + \underbrace{c_2\,\mathrm{nnz}^{\alpha}}_{\text{factorization}} + \underbrace{c_3\,n_{\mathrm{it}}\,\mathrm{nnz}_L}_{\text{iterations}} + \underbrace{c_4\,n_f k^2}_{\text{overhead}}, \tag{24}$$

while the dense route remains $T_{\mathrm{dense}} \sim n_f^3$ with storage $\mathcal{O}(n_f^2)$. Because the factorization term dominates in our range of n_f, the measured ratio $T_{\mathrm{dense}}/T_{\mathrm{sparse}}$ grows faster than n_f but we do not claim the clean n_f^2 law that a purely linear sparse cost would imply. What the measurements in Section 8.3 establish is the *behaviour*, not the exponent: the dense cost is observed to grow as n_f^3 while the sparse cost stays below a second across the two orders of magnitude in n_f.

A reproducible cost report should distinguish at least the assembly time, the factorization time and memory, the number of Arnoldi iterations, the eigensolver tolerance and the residual attained. We report the first and third in Section 8.3, and we note that the factorization is performed inside the external sparse solver and is not instrumented separately here.

6 Implementation

To enable the code-to-code verification of Section 7, the formulation of Sections 2–3 was implemented *twice*, independently, in different languages and on different infrastructure.

6.1 Elmer solver module

Elmer [21] is an open-source finite element suite developed at CSC – IT Center for Science (Finland) and used principally for multiphysics problems. Its solver interface allows the addition of user modules compiled against the Elmer libraries. We implemented `SchrodingerSolver` (about 460 lines of Fortran 90) following this interface. Meshes for all cases were generated with Gmsh [28] and converted to the Elmer format with ElmerGrid. The module reads the mesh through the standard element interface; identifies constrained nodes from the boundary elements, giving the free/constrained partition of Proposition 1; reads the potential as described in Section 6.3; assembles K, M and M_V; solves the dense generalized symmetric problem with `DSYGV`; writes the ground-state eigenfunction back to the Elmer field database so that standard output modules produce VTK files; and optionally exports A_{ff} and M_{ff} in ASCII format for the external sparse route.

Documentation was written following the structure of the official Elmer manuals—model manual, solver manual and per-case tutorials—to lower the barrier for external users.

6.2 Python reference implementation

The second implementation is a dimension-generic Python code sharing no source code with the Elmer module. It comprises a mesh generator and assembler for structured 1D, 2D and 3D grids using direct Gauss quadrature as in (11); the sparse shift-and-invert Arnoldi solver of Section 5, built on ARPACK through SciPy, with the same interface as the dense solver so that the two can be interchanged; and a PETSc/SLEPc driver implementing distributed Krylov–Schur with shift-and-invert, together with conversion utilities for the assembled matrices. The Python code is the one used for the performance measurements, since it allows the dense and sparse strategies to be exercised on identical matrices.

6.3 Potential specification in a general framework

Specifying a multidimensional potential in a general finite element code requires care, and we document a practical obstacle. The expression parser of Elmer does not expose the individual coordinate components in a form that allows a two- or three-dimensional expression to be written directly for the potential field; the syntaxes we tested—a coordinate list, per-component indexing, and separate named variables—all failed with parser errors.

Table 1 Analytic reference solutions used for verification, in the unit convention of (3).

potential	notation	eigenvalues	eigenfunctions
infinite well (1D)	$V = 0$, $\psi(\pm L/2) = 0$	$E_n = (n\pi/L)^2$	trigonometric
infinite box (dD)	$V = 0$, $\psi\|_\partial = 0$	$E = \pi^2 \sum_k n_k^2/L^2$	products
harmonic, $V = \alpha r^2$	isotropic	$E_n = (2n + d)\sqrt{\alpha}$	$H_n\times$ Gaussian
Morse	$D_e(1 - e^{-a(x-x_0)})^2 - D_e$	closed form	—

We therefore implemented a component-wise convention *inside the solver*: the potential is read as a sum of independent nodal fields, one per coordinate,

$$V(x_1, \ldots, x_d) = \sum_{k=1}^{d} V_k(x_k), \tag{25}$$

each obtained by an independent database query, with the precedence order: (i) a directly specified nodal field `Potential` (arbitrary V, used in 1D); (ii) component fields `Potential X`, `Potential Y`, `Potential Z`, summed by the solver (used in 2D and 3D); (iii) $V \equiv 0$ if none is present. The restriction to separable potentials is mild for the verification suite considered here; removing it would require extending the expression interface rather than the solver core, and we return to this in Section 9.2.

7 Verification

Verification—establishing that the code solves the equations it is intended to solve—is treated here as a first-class component. It is to be distinguished from validation, which establishes agreement with experiment; for an experimental validation study in a comparable electromagnetic setting, see [29]. Four independent checks are applied here, following the philosophy of the verification literature [16, 17].

7.1 Analytic benchmarks

Closed-form solutions are available for the potentials in Table 1 (Appendix A); they are classical [2, 4]. The infinite well is a genuine test of the discretization: the eigenvalue problem is satisfied in the interior with homogeneous boundary data, and the eigenvalues are known exactly.

7.2 Normalisation of the analytic ground state

Because the Gaussian of (A2) is used as a reference in the tests above, its prefactor was checked numerically against $\int |\psi_0|^2 \mathrm{d}^d r = 1$ for several values of α. Writing the prefactor as $c(\alpha)^d$ with $c = (\sqrt{\alpha}/\pi)^{1/4}$, the product form of the integral gives $c^{2d}(\pi/\sqrt{\alpha})^{d/2} = 1$ for every α. Using the incorrect $(\alpha/\pi)^{d/4}$ instead yields a total norm of 0.500 for $\alpha = 1/4$ in 2D and 0.354 in 3D, and the two prefactors coincide only at $\alpha = 1$. This is why the 1D case plotted in Figure 2, which uses $\alpha = 1$, does not detect the error, whereas the 2D and 3D benchmark cases with $\alpha = 1/4$ do.

Table 2 Measured convergence of the first eigenvalue. The fitted order is obtained from consecutive refinements and compared with the theoretical value $2s = 2$ of Corollary 3.

discretization	h	relative error in E_1	fitted order
1D well, $N = 25$	1.600	1.317×10^{-3}	—
1D well, $N = 50$	0.800	3.290×10^{-4}	2.00
1D well, $N = 100$	0.400	8.225×10^{-5}	2.00
1D well, $N = 200$	0.200	2.056×10^{-5}	2.00
1D well, $N = 400$	0.100	5.140×10^{-6}	2.00
2D box, $n = 10$	1.000	8.251×10^{-3}	—
2D box, $n = 20$	0.500	2.058×10^{-3}	2.00
2D box, $n = 40$	0.250	5.141×10^{-4}	2.00
2D box, $n = 80$	0.125	1.285×10^{-4}	1.99

7.3 Convergence order

Corollary 3 predicts $\mathcal{O}(h^2)$ for the first eigenvalue on convex domains with multilinear elements. We measured this directly. For the 1D infinite well with $L = 40$, refining from $N = 25$ to $N = 400$ elements, the relative error in E_1 falls from 1.317×10^{-3} to 5.140×10^{-6}, giving a fitted order of 2.00 over four refinements. In 2D, refining a 10×10 to an 80×80 mesh for the square well reduces the error from 8.251×10^{-3} to 1.285×10^{-4}, also order 2.00. Table 2 gives both sequences and Figure 1(a) shows them against the reference slope. The measured orders agree with (19), which upgrades the observed convergence from an empirical observation to a predicted-and-confirmed property.

Throughout this section h denotes the diameter of an element, so that for the uniform structured meshes used here $h = L/n$ in one dimension and $h = L/n$ in each coordinate direction in higher dimensions, with n the number of elements per side. Each refinement halves h exactly, which makes the fitted order $\log(e_i/e_{i+1})/\log 2$ rather than an approximation; e_i is the relative error $|E_{h,1} - E_1|/|E_1|$ against the analytic value of Table 1. All meshes are uniform, so the shape-regularity hypothesis of Theorem 2 is satisfied with a constant that does not degrade under refinement.

Table 2 concerns the eigenvalues. The eigenfunctions converge more slowly, and Table 3 reports their error for the same 1D well, measured against the exact $\psi(x) = \sqrt{2/L}\cos(\pi x/L)$ in the L^2 and H^1 norms (sign aligned). The H^1 error decreases at order 1 while the L^2 error decreases at order 2, as expected from the usual Aubin–Nitsche duality argument applied on top of Theorem 2 (which states the energy-norm estimate and the eigenvalue estimate directly): the energy-norm error is $\mathcal{O}(h^s)$ with $s = 1$, one order worse than the eigenvalue error, while the additional order in L^2 is the usual Aubin–Nitsche gain. This is the practical content of the doubling of the order mentioned after Theorem 2.

7.4 Code-to-code verification

The strongest evidence in this study comes from comparing the two independent implementations. Using identical meshes and potentials, and requesting the lowest

Table 3 Error of the first *eigenfunction* for the 1D infinite well ($L = 40$), against the exact solution, after aligning the sign. The orders are computed as in Remark 6.

n	h	$\|\psi_h - \psi\|_{L^2}$	order	$\|\psi_h - \psi\|_{H^1}$	order
25	1.600	5.90×10^{-4}	—	2.91×10^{-3}	—
50	0.800	1.47×10^{-4}	2.00	1.43×10^{-3}	1.02
100	0.400	3.68×10^{-5}	2.00	7.13×10^{-4}	1.01
200	0.200	9.20×10^{-6}	2.00	3.56×10^{-4}	1.00
400	0.100	2.30×10^{-6}	2.00	1.78×10^{-4}	1.00

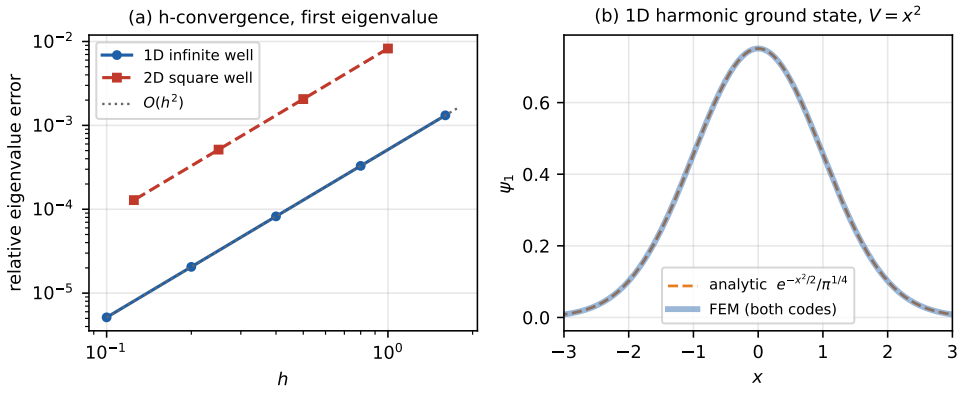


Fig. 1 (a) Relative error of the first eigenvalue against mesh size h for the 1D infinite well and the 2D square well, with the $\mathcal{O}(h^2)$ reference slope predicted by Corollary 3. (b) Ground state of the 1D harmonic oscillator, $V = x^2$: both implementations produce the analytic Gaussian to within the line width.

eigenvalue, the Elmer module and the Python code agree to within 4×10^{-13} in relative terms across all three spatial dimensions (Table 4). The last column of that table gives the normalised residual $\|A_{ff}\boldsymbol{\psi} - EM_{ff}\boldsymbol{\psi}\|/\|A_{ff}\boldsymbol{\psi}\|$ of the computed eigenpair. Both norms are Euclidean and the eigenvector is normalised by $\boldsymbol{\psi}^\top M_{ff}\boldsymbol{\psi} = 1$; dividing by $\|A_{ff}\boldsymbol{\psi}\|$ makes the quantity dimensionless and independent of the scaling of A, which is what allows rows with eigenvalues of very different magnitude to be compared.

The level of agreement should be read with the conditioning of the problem in mind. For the 1D well the first eigenvalue, 6.2×10^{-3}, is three orders of magnitude smaller than $\|A_{ff}\| \approx 1/h = 6.25$, so the relative accuracy attainable in E_1 is limited to roughly $\varepsilon\,\|A_{ff}\|/E_1 \approx 10^{-12}$ in double precision, whatever algorithm is used; the residual of 3.2×10^{-11} in that row reflects the same conditioning. For the 2D and 3D cases the eigenvector is well conditioned and both the eigenvalue difference and the residual are at the 10^{-13} and 10^{-14} levels respectively. We checked that three independent solution paths for the 1D case (the in-framework dense solve, the exported sparse solve, and an independent dense implementation written for this verification)

Table 4 Code-to-code verification: two independently written implementations on identical meshes and potentials.

case	Elmer E_1	Python E_1	relative difference	$\|A\psi - EM\psi\|/\|A\psi\|$
1D infinite well	$6.168583925347 \times 10^{-3}$	$6.168583925345 \times 10^{-3}$	3.6×10^{-13}	3.2×10^{-11}
2D harmonic	1.028585384356000	1.028585384356285	2.8×10^{-13}	4.8×10^{-14}
3D harmonic	1.625507320678000	1.625507320677697	1.9×10^{-13}	1.1×10^{-14}

Table 5 The exported-matrix sparse route reproduces the in-framework dense solution to all displayed digits.

route	E_1	E_2	E_3	E_4
in-framework (dense `DSYGV`)	1.113644656498	2.172208383659	2.172208383659	3.230772110820
exported + sparse (ARPACK)	1.113644656498	2.172208383659	2.172208383659	3.230772110820

differ among themselves by up to 5×10^{-12} in relative terms, which is the same round-off scale.

Because the two codes share no source code, use different mesh containers, different assembly loops and different eigensolver libraries, agreement at this level strongly constrains a broad class of implementation errors that comparison against analytic solutions alone would not detect. The approach follows the spirit of earlier cross-validation of independent codes in computational physics [20] and is, in our assessment, the single most informative result of the verification programme—available precisely because two implementations were developed.

A related internal check is that solving the *exported* matrices with the sparse pipeline reproduces the dense in-framework solution (Table 5). The twelve digits shown coincide, so the export introduces no error at the level of the reported precision and the sparse route is a drop-in replacement for the dense one. This is a statement about the two solution paths agreeing with each other, not about either being exact: both solve the same discrete problem, and both inherit whatever discretization error that problem carries.

7.5 Structural consistency checks

Three checks require no external reference.

Mass-orthonormality. The computed eigenvectors must satisfy $\boldsymbol{\psi}_i^\top M \boldsymbol{\psi}_j = \delta_{ij}$. We verified this numerically; in the 2D case the largest deviation from the identity is 3.1×10^{-15}, that is, at the level of double-precision round-off.

Degeneracies. Symmetry of the domain and potential implies degeneracies of known multiplicity. The 2D isotropic harmonic oscillator on a square domain must have a doubly degenerate first excited state; we obtain $E_2 = E_3 = 2.043966018888$ to all displayed digits. In 3D the first excited state is triply degenerate; we obtain

$E_2 = E_3 = E_4 = 2.669307913349$. Degeneracies are a strong test because they are destroyed if the assembly breaks the symmetry—for instance through inconsistent element ordering or a dimension-dependent quadrature error.

Virial theorem. For a harmonic potential the virial theorem [2] requires $\langle T\rangle = \langle V\rangle = E/2$. Multiplying (3) by ψ and integrating by parts gives $\langle T\rangle = \int |\nabla\psi|^2 \mathrm{d}\Omega$ and $\langle V\rangle = \int V|\psi|^2 \mathrm{d}\Omega$, so that $E = \langle T\rangle + \langle V\rangle$ holds identically. Discretely, the natural definitions are the quadratic forms of the assembled matrices, $\langle T\rangle_h = \boldsymbol{\psi}^\top K \boldsymbol{\psi}$ and $\langle V\rangle_h = \boldsymbol{\psi}^\top M_V \boldsymbol{\psi}$ with $\boldsymbol{\psi}$ normalised by $\boldsymbol{\psi}^\top M \boldsymbol{\psi} = 1$; then $E_h = \langle T\rangle_h + \langle V\rangle_h$ holds exactly by construction, and only the ratio is a test. For the 1D harmonic ground state we obtain

n	$\langle T\rangle_h$	$\langle V\rangle_h$	$\langle T\rangle_h / \langle V\rangle_h$
100	0.5000031	0.5091573	0.98202
200	0.5000002	0.5022911	0.99544
400	0.5000000	0.5005729	0.99886
800	0.5000000	0.5001432	0.99971
1600	0.5000000	0.5000358	0.99993

The ratio approaches the exact value 1 as $\mathcal{O}(h^2)$, while $\langle T\rangle_h$ is already accurate to 10^{-8} at $n = 400$. The check is notable because it relates the eigenvalue to the eigenfunction without invoking the analytic form of the latter.

Two points about the finite box are worth separating. The identity $E = \langle T\rangle + \langle V\rangle$ is an integration by parts whose boundary term is $\psi\, \partial_n \psi$ on $\partial\Omega$, which vanishes because $\psi = 0$ there; it therefore holds in a bounded box with no correction, and the table confirms it ($E_h - \langle T\rangle_h - \langle V\rangle_h$ is at round-off, $\sim 10^{-14}$). The ratio $\langle T\rangle_h / \langle V\rangle_h$ is a different statement, derived from the scaling $x \mapsto \lambda x$, which in a bounded box is not a symmetry of the problem. We checked numerically that it is nevertheless independent of the box: at fixed $h = 0.25$ and $\alpha = 1$ the ratio equals 0.97221 for every L from 12 to 28, so the deviation from 1 is entirely the $\mathcal{O}(h^2)$ discretization error above and not a truncation effect. The first row of the table, 0.98202 at $n = 100$, and the value 0.97221 at $h = 0.25$ are consistent with this reading.

Monotonicity. By Theorem 1 the computed eigenvalues must approach the exact ones from above under refinement. All sequences in Table 2 satisfy this.

8 Numerical experiments

8.1 Benchmark suite

Unless stated otherwise, the timings and residuals reported below were obtained on a single workstation with an Intel i7-1065G7 processor (4 physical cores, 8 logical) and 31 GB of RAM, running Linux with Elmer 9.0 built against GFORTRAN 9.4.0, MKL and MUMPS, Python 3.10 with NUMPY 1.24.4 and SCIPY 1.10.1. The eigensolver tolerances are those of the libraries: LAPACK `DSYGV` uses its default, ARPACK is called with `tol` $= 0$, so its stopping criterion is machine precision, and the Elmer module uses the default of its own solver interface. The distributed runs of Section 8.4

Table 6 Benchmark suite: four lowest eigenvalues for each case, in the unit convention of (3). The last column gives the number of free degrees of freedom n_f after elimination (Proposition 1).

case	E_1	E_2	E_3	E_4	n_f
1D infinite well	6.168584×10^{-3}	2.467531×10^{-2}	5.552310×10^{-2}	9.871683×10^{-2}	249
1D harmonic	1.000572892	3.001197697	5.002447062	7.004320695	399
1D Morse	−8.481033787	−5.818565432	−3.656064809	−1.991029948	399
1D double well	0.224110514	0.600028005	1.274687373	2.059803060	399
1D sinusoidal	−0.213200584	−0.209442228	0.285067468	0.343455929	399
2D harmonic	1.028585384	2.043966019	2.043966019	3.059346653	1521
3D harmonic	1.625507321	2.669307913	2.669307913	2.669307913	2197

were made on a cluster node with 256 cores and 251 GB of RAM. Timings are the wall-clock time of the eigensolution call alone, excluding assembly; the sparse figures are the minimum of three repetitions, and the dense figures a single run because of their cost. Absolute times are properties of the hardware and are not transferable; the algorithmic comparison is.

Seven cases were run to completion with the Elmer module, covering smooth and non-smooth potentials, bound and quasi-bound states, and one- and multidimensional geometries. Table 6 lists the four lowest eigenvalues of each, together with the number of free degrees of freedom. Figure 2 shows the potentials and the computed levels.

Two features merit comment.

The double well. With a barrier separating two symmetric minima, the two lowest states are split by tunnelling; we obtain $E_2 - E_1 = 0.376$, a splitting that vanishes if the barrier is removed. The presence of a *finite* splitting is a sensitive indicator that the barrier region is resolved and that the two wells are correctly coupled through it: a mesh too coarse in the barrier overestimates it, and a symmetry-broken discretization can suppress it artificially. Double-well structures of this kind are the standard model for coupled quantum wells and for the tunnelling phenomena exploited in resonant tunnelling devices [9].

The periodic potential. The lowest two eigenvalues, -0.213200584 and -0.209442228, lie just above the potential minimum $V_{\min} = -0.5$ and are separated by only 3.8×10^{-3}, a near degeneracy that is a property of this bounded interval and not, by itself, a band structure: the computation imposes Dirichlet conditions and no quasi-momentum is defined, so the levels are those of a finite segment of a periodic potential rather than of a crystal. Establishing a band structure would require Bloch conditions or a superlattice with many periods, and is outside the present scope. Potentials of this type model superlattices and miniband structures [4]; for the computation of genuine band structures, where the eigenproblem is again of the shift-and-invert type, see [30].

8.2 Discretization error versus box truncation

The harmonic cases deserve care, because two distinct error sources are easily conflated there. The analytic eigenvalues $E_n = n + d/2$ in the convention of (3) refer to an unbounded domain, whereas the computation imposes $\psi = 0$ on the boundary of

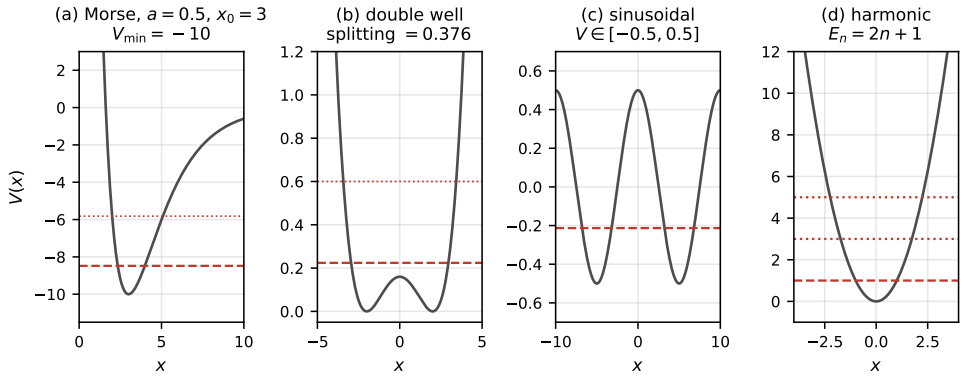


Fig. 2 Potentials of the 1D benchmark cases with the computed levels (dashed and dotted lines): (a) Morse, $V_{\min} = -10$; (b) double well, splitting $E_2 - E_1 = 0.376$; (c) sinusoidal, $V \in [-0.5, 0.5]$; (d) harmonic, $E_n = 2n + 1$.

a finite box of side L. Both the finite box and the finite mesh raise the computed eigenvalue, but by very different amounts; in the cases considered here it is the mesh that dominates.

The truncation effect is negligible at these box sizes. By (A2) the analytic ground state is $\exp(-\sqrt{\alpha}\, r^2/2)$, so writing it as $\exp(-r^2/2\sigma^2)$ the characteristic length is $\sigma = \alpha^{-1/4}$—which for $\alpha = 1/4$ is $\sqrt{2}$ and *not* $\sqrt{\alpha} = 1/2$. On a centred box of side L the amplitude at a face is $\exp(-\sqrt{\alpha}\, L^2/8)$; for $\alpha = 1/4$ and $L = 12$ that is $\exp(-9) \approx 1.2 \times 10^{-4}$ of the peak, so the exact solution very nearly satisfies the Dirichlet condition already. *If* one assumes the eigenvalue perturbation to behave like the square of that amplitude, the estimate is 10^{-8} or less; this is an order-of-magnitude heuristic and not a demonstrated bound. The direct evidence is in Table 7, where the box is varied at fixed mesh size.

That the discrepancy is instead a mesh effect is confirmed by refinement. Table 8 refines the 2D and 3D harmonic cases at fixed box size. The errors decrease as $\mathcal{O}(h^2)$, in agreement with Corollary 3, and the sequences converge to the *unbounded* values 1.0 and 1.5 rather than to any larger box eigenvalue. Refining the mesh at fixed L therefore recovers the analytic result, which would not occur if the box were the controlling error.

The discrepancies quoted in Table 6—2.9% in 2D at $h = 0.5$ and 8.4% in 3D at $h = 0.857$—are thus discretization error at a deliberately coarse mesh. The distinction is not merely semantic: reducing them requires refining the mesh, not enlarging the box. Higher excited states, which oscillate faster, are affected more strongly, and errors of 20–40% appear in the fourth eigenvalue at these mesh sizes. Box dimensions and mesh sizes are therefore reported explicitly (Appendix B).

A second, independent check separates the two effects by varying the box at *fixed* mesh size. Table 7 reports the first eigenvalue of the 2D and 3D harmonic cases for a sequence of boxes, with h held at 0.5 and 1.0 respectively. In 2D the value is constant to twelve significant digits from $L = 16$ onwards. In 3D, where the fixed

Table 7 Domain-size study at *fixed* mesh size. Once the box is large enough the first eigenvalue stops changing, while the offset from the unbounded value does not: the offset is discretization, not truncation. The variations in the last digits are round-off.

2D harmonic, $h = 0.5$, $\alpha = 1/4$, unbounded $E_1 = 1$				
L	12	16	28	40
n_f	529	961	3 025	6 241
E_1	1.028585446	1.028585384	1.028585384	1.028585384
3D harmonic, $h = 1.0$, $\alpha = 1/4$, unbounded $E_1 = 1.5$				
L	6	9	12	18
n_f	125	512	1 331	4 913
E_1	1.737955701	1.670595303	1.670466988	1.670466985
ΔE_1	—	6.7×10^{-2}	1.3×10^{-4}	3×10^{-9}

Table 8 Refinement of the harmonic cases at fixed box size. Both sequences converge to the *unbounded* analytic values rather than to a larger box eigenvalue, showing that the leading error is discretization and not box truncation. The last column gives the order observed between consecutive refinements, $\log(e_i/e_{i+1})/\log(h_i/h_{i+1})$, computed from the relative errors e_i (Remark 6).

case	n	h	E_1	relative error	observed order
2D, $L = 20$, $\alpha = 1/4$	20	1.000	1.113645	1.14×10^{-1}	—
	40	0.500	1.028585	2.86×10^{-2}	1.99
	60	0.333	1.012719	1.27×10^{-2}	2.00
	80	0.250	1.007158	7.16×10^{-3}	2.00
3D, $L = 12$, $\alpha = 1/4$	10	1.200	1.744630	1.63×10^{-1}	—
	14	0.857	1.625507	8.37×10^{-2}	1.98
	18	0.667	1.576105	5.07×10^{-2}	1.99
	20	0.600	1.561688	4.11×10^{-2}	1.99

mesh is much coarser, it is constant to nine significant digits from $L = 12$ onwards; the coarser plateau reflects the larger discretization error and not a larger truncation effect, since the box is already several times the characteristic length of the ground state. In both cases the offset from the unbounded analytic value (1 and 1.5) is a pure discretization error at the fixed mesh size—2.9×10^{-2} at $h = 0.5$ in 2D and 1.7×10^{-1} at $h = 1.0$ in 3D—and does not shrink as the box grows. Truncation is therefore below the resolution of the computation over the range tested, which is the direct evidence that the discrepancies of Table 6 must be attributed to the mesh. The statement is limited to the boxes and mesh sizes reported: a box smaller than $L = 12$ in 3D does show a truncation error, as the first row of Table 7 makes plain.

Remark 7 (Quadrature of the potential term) The error constant depends on how the potential enters the weighted mass matrix (11). Both implementations assemble $\int N_i V_h N_j$, that is, they use the *interpolant* V_h of V in the finite element space rather than evaluating V exactly at the quadrature points—which is the natural convention in a general finite element

Table 9 Eigensolution time for the 2D harmonic case, $k = 4$, as a function of the number of free degrees of freedom n_f. Identical matrices in both columns. The last row is not merely slow in dense arithmetic but exceeds the memory of the workstation used (31 GB): at $24\,n_f^2$ it requires 37.6 GB, cf. (21). On a large-memory node it would fit, but the sparse route makes the question moot. In the first row the sparse time is below the resolution of the timer, so no speed-up is quoted there and none is annotated in Figure 3.

n_f	dense `DSYGV` (s)	sparse ARPACK (s)	speed-up	dense memory
361	0.01	< 0.01	—	3.1 MB
1 521	0.58	0.01	58×	55.5 MB
6 241	41.06	0.05	821×	935 MB
14 161	491.60	0.12	4097×	4.81 GB
39 601	infeasible (37.6 GB)	0.41	—	37.6 GB

code, where the potential is supplied as a nodal field. For the quadratic potentials considered here this introduces an additional $\mathcal{O}(h^2)$ contribution; it does not change the order of convergence but does increase the constant. Recomputing the 1D harmonic case ($L = 20$, 400 elements) with V evaluated exactly at the quadrature points raises the accuracy of the first eigenvalue from 1.000572892 to 1.000156243, against the exact value 1. We retain the interpolant convention and record the difference here.

8.3 Eigensolution performance

Table 9 and Figure 3 compare the dense and sparse strategies on the 2D harmonic case, holding the mesh and the number of requested eigenpairs ($k = 4$) fixed. The discretization is identical in all rows; only the solver changes. The first column gives the dimension of the reduced matrices, that is, the number of free degrees of freedom after the elimination of Proposition 1; for a structured mesh of $n \times n$ elements on a square with Dirichlet data on the whole boundary this is $(n - 1)^2$. The measurements were taken on a workstation with 31 GB of memory, and were reproduced on a cluster node; the absolute times differ between the two machines, but the algorithmic behaviour is the same.

The measured speed-up grows faster than linearly. A factor 70 in speed-up for a factor 9.3 in problem size would be consistent with an $\mathcal{O}(n_f^2)$ ratio, but with three interior points we do not claim to have identified an asymptotic exponent; what the data establish is the behaviour—sub-quadratic cost in n_f against cubic for the dense route—and the resulting crossover, not a fitted law $T_{\text{dense}}/T_{\text{sparse}}$. The comparison at the largest case is not a timing comparison at all: the dense route cannot be run. This is the practical argument for the sparse strategy—it is not merely faster, it extends the accessible problem class.

Thread-level parallelism does not substitute for the algorithmic change. Enabling multithreaded BLAS in the sparse solve at $n_f = 39601$ we obtain 0.84 s with one thread, and 1.10, 1.10 and 2.25 s with two, four and eight threads respectively (minimum of three repetitions). Threading therefore gives no speed-up at all here—it degrades the result, since the extra workers contend for the same memory bandwidth

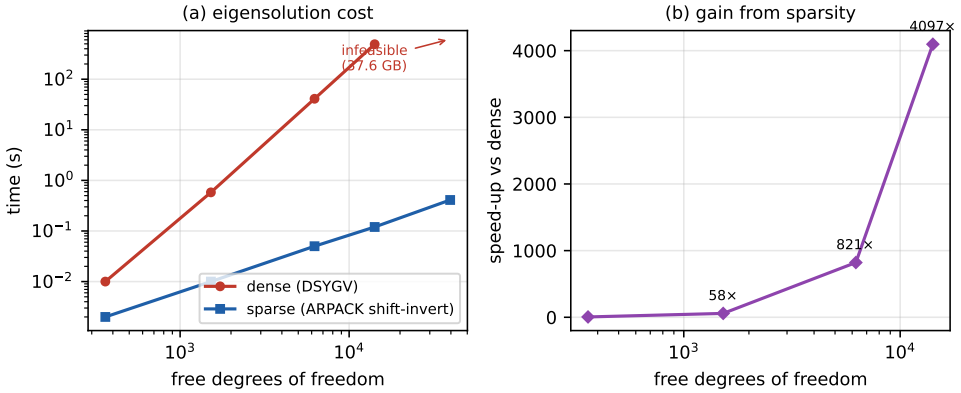


Fig. 3 (a) Eigensolution time against problem size for the dense and sparse strategies, on identical matrices. (b) Speed-up of the sparse strategy over the dense one, with measured values annotated.

while the sparse factorization itself remains sequential. This reinforces the point of Section 5: the gain must come from the algorithm, not from the hardware.

8.4 Behaviour under domain decomposition

We report a negative result, because it defines the boundary of applicability of the in-framework solver and motivates the second route discussed in Section 9.1.

The Elmer module is *not* MPI-aware, and two distinct problems arise when it is run under `mpirun -np` with more than one process.

The first is a defect in Elmer 9.0 independent of our module. The routine that builds the mesh path for a partitioned run issues only a *warning*—not a fatal error—when the partition header file is absent, and returns a null pointer. The caller dereferences the result *before* the associated-pointer check a few lines later, producing a segmentation fault. A controlled A/B test confirms the diagnosis: the same binary segfaults at `-np 2` without the partition directory and exits cleanly with it present.

The second problem is specific to the solver and more serious. Even when the mesh *is* partitioned, the module assembles dense *local* matrices on each rank. The assembly loop uses rank-local element and node counts, so each rank solves a different, smaller problem and the results are simply wrong. On a small test mesh, where the serial result can be checked by hand, we obtain:

	serial	`-np 2`
free degrees of freedom	59	29 *per rank*
E_1	1.025415149101	3.052806819296

The failure is silent—the run completes and reports eigenvalues—which makes it more dangerous than a crash. We therefore do not recommend distributed execution of the in-framework solver in its present form. The correct fix, a solver with native parallel

assembly following the pattern of the existing Elmer wavefunction solver, remains future work.

9 Discussion

9.1 Two routes and their crossover

The results above suggest a practical division of labour rather than a single recommended strategy.

In-framework route (dense, serial). Up to roughly 10^4 free degrees of freedom—which covers a wide range of one- and two-dimensional quantum-well and quantum-dot calculations at engineering accuracy—the dense in-framework solver is the simpler choice. It requires no matrix export and keeps the entire workflow inside the finite element framework, including post-processing. The $n_f = 6561$ case solves in 41 s, acceptable for interactive use.

Export route (sparse, parallel-capable). Beyond that size, and whenever several eigenpairs are needed or the mesh must be refined, the matrices should be exported and solved externally. This route reproduces the dense result exactly (Table 5), reduces storage from $\mathcal{O}(n_f^2)$ to $\mathcal{O}(\mathrm{nnz})$, and—given a parallel sparse direct factorization—scales across nodes. The exported matrices are the *same* matrices: the route is a change of linear algebra, not of discretization, so the two solve the same discrete problem and agree to the reported precision (Table 5). They are not required to agree bit for bit, since two different algorithms on the same matrices need not round identically; the choice between them is one of computational resources.

The crossover follows from (21), and the factor matters. If only the irreducible two matrices are retained, $16n_f^2$ bytes are needed and the dense route becomes impractical at $n_f \sim 10^4$ for a 1.6 GB budget and $n_f \sim 4 \times 10^4$ for 25 GB. The implementation measured here holds three matrices, so the relevant coefficient is $24n_f^2$ and the same budgets are reached earlier, at $n_f \sim 8 \times 10^3$ and $n_f \sim 3.2 \times 10^4$. Both figures are estimates from the storage model alone; they take no account of the workspace the eigensolver itself allocates, which only makes the dense route less attractive.

9.2 Limitations

Separable potentials. The component-wise convention of Section 6.3 restricts the module to potentials of the form $\sum_k V_k(x_k)$. Non-separable potentials—a Coulomb term, or the $1/r$ inter-electronic interaction used in quantum-dot models [5]—would require extending the expression interface, not the solver core. Coupling to a self-consistent Poisson equation, as in the Schrödinger–Poisson problem [9, 31], is likewise outside the present scope, as is the treatment of open boundary conditions and polar geometries [32]. For time-dependent and nonlinear formulations, which lie outside the scope considered here, we refer to [33, 34].

No parallel in-framework solver. As documented in Section 8.4, the module is serial; parallelism is obtained through the export route. The scaling of that route is limited by the availability of a parallel sparse direct factorization in the deployed PETSc build; on the installation used here, PETSc was configured without a parallel

LU, so the shift-and-invert step could not be distributed and the driver was confined to one process. The driver is complete and verified at one process; only the linear-algebra backend is missing. We regard this as a deployment issue rather than a methodological limitation, but it does mean we cannot report parallel scaling numbers, and we prefer to say so.

Coarse meshes for the confined cases. As shown in Section 8.2, the discrepancies in the harmonic cases are discretization error, not box truncation: the boxes are large enough that the truncation contribution is several orders of magnitude smaller. Reducing those discrepancies therefore requires mesh refinement, and the meshes used for the 2D and 3D harmonic cases are deliberately coarse. For potentials that decay slowly, or for boxes chosen too small, truncation would become the leading error and would then have to be separated from the discretization error.

Only the ground state is written to file. The module returns all requested eigenvalues but writes only the first eigenfunction to the field database, so the standard output modules produce a single mode per run.

9.3 On the nature of the contribution

We state plainly that the work introduces no new numerical algorithm. The elements are standard multilinear Lagrange elements, the quadrature is standard Gauss–Legendre, the eigensolvers are LAPACK, ARPACK and SLEPc, and the weak form is textbook. Because the novelty is not algorithmic, we prefer to state explicitly what the paper does contribute, so that the reader can weigh each item against the alternatives.

Reusable software. A dimension-generic solver for (1) distributed as a module of Elmer, with all seven benchmark cases, their meshes, their input files and a container definition that reproduces the environment. The module did not previously exist in that framework, and the packaging is intended so that a third party can reproduce every number in this paper without contacting the authors.

Corrected analysis for a common pitfall. Section 3.3 shows, with numerical evidence, that the diagonal-shift treatment of Dirichlet data used routinely for source problems is invalid here, and distinguishes the two variants of that recipe and their different failure modes. The proof of Proposition 1 is written in terms of the variational restriction, which is the form that survives scrutiny; the naive block-matrix argument, as Remark 3 shows, happens to hold for exponentially decaying eigenfunctions and fails for the infinite well.

Scoped theoretical statements. Theorems 1 and 2 are stated with the hypotheses they actually require. Remark 4 records that the upper-bound property is not preserved by the nodal interpolation of the potential for non-convex V, and that it is therefore verified numerically rather than assumed; Remark 6 separates the theoretical convergence order from the order observed on a particular sequence of meshes.

Quantified comparison. Dense and sparse eigensolution strategies are compared on identical matrices, with a cost model that separates assembly, factorization, iteration and solver overhead (Section 5.4), and the crossover between the two routes is estimated. The distributed route is analysed and its present limitations documented; it is not benchmarked, and we say so.

A verification programme. Four independent checks—analytic benchmarks, code-to-code comparison of two independently written implementations, structural invariants, and convergence orders—each reporting residuals and conditioning where relevant. We consider the code-to-code comparison the strongest of the four, and Section 7.4 explains why its apparent agreement is limited by the conditioning of the problem rather than by the quality of either code.

We believe this profile is representative of a useful class of work in computational science: not new mathematics, but verified, characterised and reusable numerical software, accompanied by analysis that is correct as stated rather than correct by accident.

10 Conclusions

We have presented a finite element framework for the stationary Schrödinger eigenvalue problem, implemented independently in two codes and verified in four ways. The main findings are the following.

1. The generalized symmetric eigenproblem $A\psi = EM\psi$ with $A = K + M_V$ is solved accurately by multilinear finite elements in one, two and three dimensions. Where the discrete form is the exact restriction of the continuous one, the conforming approximation yields eigenvalues that are upper bounds for the exact ones (Theorem 1); the implementation's nodal interpolation of the potential preserves that property for convex V and is checked numerically for the others (Remark 4). The eigenvalue error obeys the classical $\mathcal{O}(h^{2s})$ estimate of Theorem 2; the predicted $\mathcal{O}(h^2)$ order is confirmed numerically over four refinements in 1D and four in 2D.
2. Elimination of the constrained degrees of freedom reproduces the constrained spectrum exactly (Proposition 1), whereas the common diagonal-shift treatment introduces spurious eigenvalues at $E = 1$.
3. Two independently written implementations—a Fortran module for Elmer and a Python reference code—agree to approximately 4×10^{-13} on identical problems. This code-to-code comparison is the strongest single verification result available and is strictly more informative than comparison against analytic solutions alone.
4. The choice of eigensolution strategy dominates the computational cost. Replacing a dense solve by sparse shift-and-invert Arnoldi iteration yields speed-ups from $58\times$ to $4097\times$ and reduces storage from $\mathcal{O}(n_f^2)$ to $\mathcal{O}(\mathrm{nnz})$, extending the accessible problem size from roughly 10^4 to at least 4×10^4 degrees of freedom on the same hardware, in agreement with the cost model (24).
5. The in-framework solver is not suitable for distributed execution in its current form: two distinct failure modes were identified and are documented here. Exporting the matrices and solving them with a sparse distributed library can be made correct; whether it is faster on a given problem depends on the availability of a parallel sparse direct solver, which we did not have and therefore did not measure.
6. Structural checks requiring no external reference—expected degeneracies, mass-orthonormality, the virial theorem, and monotonicity of the eigenvalues under

refinement—are inexpensive and detect errors that analytic comparisons at a single resolution can miss.

Future work will address native parallel assembly in the solver module, the extension to non-separable and self-consistent potentials, and the integration of the solver into a multiphysics environment where the potential itself is produced by a coupled transport or electromagnetic calculation. Coupled finite element formulations of exactly this kind, in which fluid and electromagnetic fields are solved within a single model, have been developed for magnetohydrodynamic flows [35]; adding a quantum-mechanical eigenproblem to such a framework is the natural next step.

Supplementary information. The following material accompanies this article: the complete source of the Elmer solver module and of the Python reference implementation; the seven benchmark cases with their mesh definition files; step-by-step tutorials for each case following the structure of the official Elmer documentation; and a container definition that reproduces the software environment used for all reported results.

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- Consent for publication: Not applicable.
- Data availability: The mesh files and solver input files for all seven benchmark cases are included as supplementary material.
- Materials availability: Not applicable.
- Code availability: Both implementations are open source, together with the case definitions, the container definition and the tutorials: *[repository URL to be inserted]*.
- Author contribution: The author is the sole contributor to this work: formulation, implementation, verification, and writing of the manuscript.

Appendix A Analytic reference solutions

All expressions are stated in the unit convention of (3), that is, with a unit coefficient on the Laplacian. The relation to physical units is given in Section 2.2.

Infinite well, one dimension. For $V = 0$ on $[-L/2, L/2]$ with $\psi(\pm L/2) = 0$, $E_n = (n\pi/L)^2$ for $n = 1, 2, \ldots$

Infinite box, d dimensions. For a d-dimensional box of side L with homogeneous Dirichlet conditions on all faces, $E_{n_1 \ldots n_d} = (\pi^2/L^2) \sum_{k=1}^{d} n_k^2$.

Isotropic harmonic oscillator. For $V = \alpha r^2$ with $r^2 = \sum_k x_k^2$ the problem separates and

$$E_N = (2N + d)\sqrt{\alpha}, \qquad N = 0, 1, 2, \ldots, \tag{A1}$$

where N is the *total* excitation number and $E_0 = d\sqrt{\alpha}$ is the ground-state energy. The normalised ground state is

$$\psi_0(\mathbf{r}) = \left(\frac{\sqrt{\alpha}}{\pi}\right)^{d/4} \exp\left(-\frac{\sqrt{\alpha}}{2}\, r^2\right), \tag{A2}$$

which follows from $\int_{\mathbb{R}} \exp(-ax^2)\mathrm{d}x = \sqrt{\pi/a}$ with $a = \sqrt{\alpha}$ applied in each coordinate. The prefactor is $(\sqrt{\alpha}/\pi)^{d/4}$ and *not* $(\alpha/\pi)^{d/4}$; the two coincide only at $\alpha = 1$, which is why the case plotted in Figure 2 does not distinguish them, whereas the 2D and 3D benchmark cases use $\alpha = 1/4$ and do.

The multiplicity of the level with total excitation N is

$$g_N = \binom{N + d - 1}{d - 1}, \tag{A3}$$

so the first excited level, $N = 1$ and $E = d + 2$ for $\alpha = 1$, has multiplicity d—doubly degenerate in 2D and triply degenerate in 3D. This is the property used as a symmetry check in Section 7.5, and it is confirmed by the 2.669307913 eigenvalue appearing three times in the 3D row of Table 6. The multiplicity $d(d + 1)/2$ belongs to the *second* excited level, $N = 2$, $E = d + 4$, and not to the first.

Morse potential. For $V(x) = D_e(1 - e^{-a(x-x_0)})^2 - D_e$ the spectrum consists of a finite number of bound states, with

$$E_\nu = 2a\sqrt{D_e}\,(\nu + \tfrac{1}{2}) - a^2(\nu + \tfrac{1}{2})^2 - D_e, \qquad \nu = 0, 1, \ldots \tag{A4}$$

for the full line. The case used here is $D_e = 10$, $a = 0.5$ and $x_0 = 3$, on the truncated domain $[0, 10]$; at $x = 0$ the potential has already reached 111.2, so the left wall is effectively infinite, while at $x = 10$ it stands at -0.595. We use this as a quantitative check (Section 7). Because (A4) holds on the full line and the computation is performed on $[0, 10]$, the comparison carries a truncation contribution as well as a discretization one. On this mesh and box the computed levels lie above the closed-form values by 3.3×10^{-4} to 3.5×10^{-3} in absolute terms. *Relative* to each level the deviations are 3.9×10^{-5}, 8.9×10^{-5}, 2.0×10^{-4} and 1.7×10^{-3} for $\nu = 0, 1, 2, 3$: the range is 4×10^{-5} to 1.7×10^{-3}, the largest value arising in the $\nu = 3$ level because it is the smallest in magnitude, not because it is the worst resolved. A Richardson extrapolation of the mesh sequence agrees with the closed-form values to 10^{-8}, so the bulk of that difference is the discretization at $n = 400$ and the truncation is the smaller part.

Appendix B Benchmark case parameters

Table B1 Parameters of the benchmark cases. All meshes are structured and were generated with Gmsh.

case	dimension	domain	potential	elements
1D infinite well	1	$[-20, 20]$	$V = 0$	250
1D harmonic	1	$[-10, 10]$	$V = x^2$	400
1D Morse	1	$[0, 10]$	$V = 10(e^{-2a(x-x_0)} - 2e^{-a(x-x_0)})$, $a = 0.5$, $x_0 = 3$	400
1D double well	1	$[-10, 10]$	$V = 0.01(x^2 - 4)^2$	400
1D sinusoidal	1	$[-10, 10]$	$V = 0.5\cos(2\pi x/10)$	400
2D harmonic	2	$[-10, 10]^2$	$V = 0.25(x^2 + y^2)$	40×40
3D harmonic	3	$[-6, 6]^3$	$V = 0.25(x^2 + y^2 + z^2)$	14^3

In all cases homogeneous Dirichlet conditions are imposed on the whole boundary. The potentials of the 2D and 3D harmonic cases are specified through the component-wise convention of Section 6.3.

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